Code No: 123CT **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year I Semester Examinations, April/May - 2023
DIGITAL LOGIC DESIGN
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART - A

(25 Marks)

1.a) Define duality property. [2]
b) Convert the following numbers
i) $(250.5)_{10} = (\ )_2$ ii)) $(673.23)_{10} = (\ )_8$ iii) $(101110.01)_2 = (\ )_8$ [3]
c) What are prime implicants and essential prime implicants? [2]
d) Implement the following function using only NOR Gates.
$F = A(B + C) + (B.C)$ [3]
e) Why does a serial adder require only one full adder? [2]
f) Can a multiplexer be used to realize a logic function? If yes in what way, it is better than the realization using gates. [3]
g) What is a ripple counter [2]
h) What is a sequential circuit? Give the classification of sequential circuit. [3]
i) What is cache memory? [2]
j) Differentiate Static RAM and Dynamic RAM. [3]

PART - B

(50 Marks)

2.a) Realize AND, OR and EXOR gates using NAND gates.
b) State and explain the DeMorgan's Theorem and Consensus Theorem. [6+4]

OR

3.a) What is the use of complements? Perform subtraction using 7's complement for the given Base-7 numbers (565)-(666).
b) Simplify A(B+C)+AB+ABC. [5+5]

4.a) Simplify the Boolean expression using K-map and implement using NAND gates
$F(A,B,C,D) = \sum m(0,2,3,8,10,11,12,14)$.
b) Derive and Implement Exclusive OR function involving three variables using only NAND function. [5+5]

OR

5.a) Obtain the simplified expression in product of sums.
$F(A,B,C,D) = \pi(0,1,2,3,4,10,11)$
b) Obtain the SOP expression for the function given below
$F(A,B,C,D) = \sum m(0,1,2,5,8,9,10)$. [5+5]

6.a) Draw and explain the logic diagram of an excess-3 to decimal decoder with active high inputs and outputs.

b) Realize the logic expression $f = \Sigma m(0,1,3,5,8,11,12,14,15)$. [5+5]

OR

7.a) Implement a full subtractor using 3-line to 8-line decoder.

b) With the help of logic diagram and truth table, explain octal to binary encoder. [5+5]

8.a) What is race around condition? Explain in detail how to eliminate race around condition.

b) With a neat diagram explain the operation of a 10-bit ring counter. [5+5]

OR

9.a) Describe the conversion of SR-Flip-Flop to JK-Flip-Flop.

b) With a neat diagram, explain the operation of Asynchronous Up/Down counter. [4+6]

10.a) Construct the PROM using the conversion from BCD code to Excess-3 code?

b) Implement the following functions using PLA.

$A(x,y,z) = \sum m(1,2,4,6)$

$B(x,y,z) = \sum m(0,1,6,7)$

$C(x,y,z) = \sum m(2,6)$ [5+5]

OR

11.a) Describe about the memory Hierarchy in terms of capacity and access time.

b) Compare different types of ROMs. [5+5]

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